Total Presentation Time is 4 min

RAMSES

R. Teyssier, J. Palafoutas, R. Geda, J. Sunseri
Y. Pan, J-Y Lee, W. Groger
Princeton University

E. Moseley
Stanford University

B. Loring, G. Barnier
NVidia










RAMSES Main Routines

HYDRO

- Godunov MUSCL-Hancock HLLC Riemann solver
- 2 kernels: cube by Bob Caddy and paper by Troels Haugbolle
- nsubgrid = 1, 2 or 4

GRAVITY

- Multigrid (MG) Poisson solver (only nsubgrid = 1 so far)
- Hierarchical multi-resolution Gauss-Seidel iteration (V-cycle)

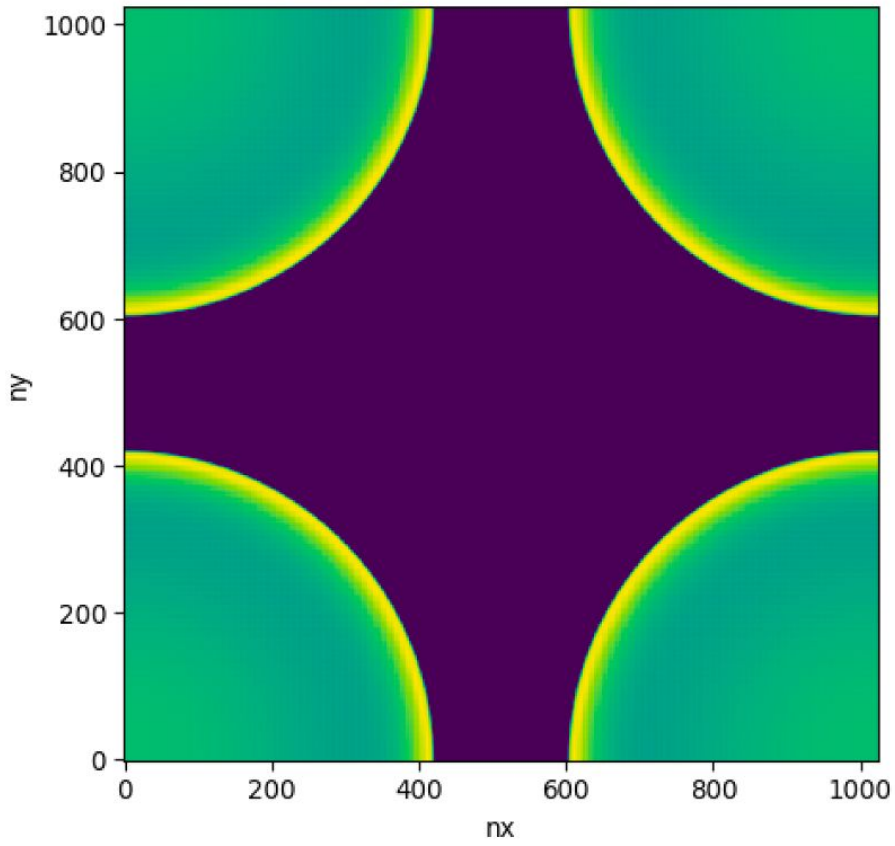
PARTICLES

- Mass deposition
- Kick-Drift-Kick pusher

AMR

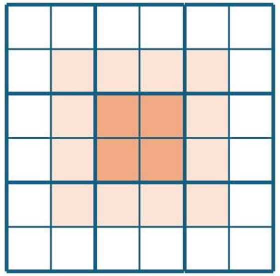
- Refinements/de-refinements
- Ghost zones

HYDRO SOLVER

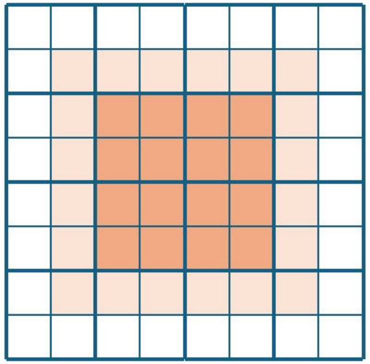


Higher performance with larger stencil

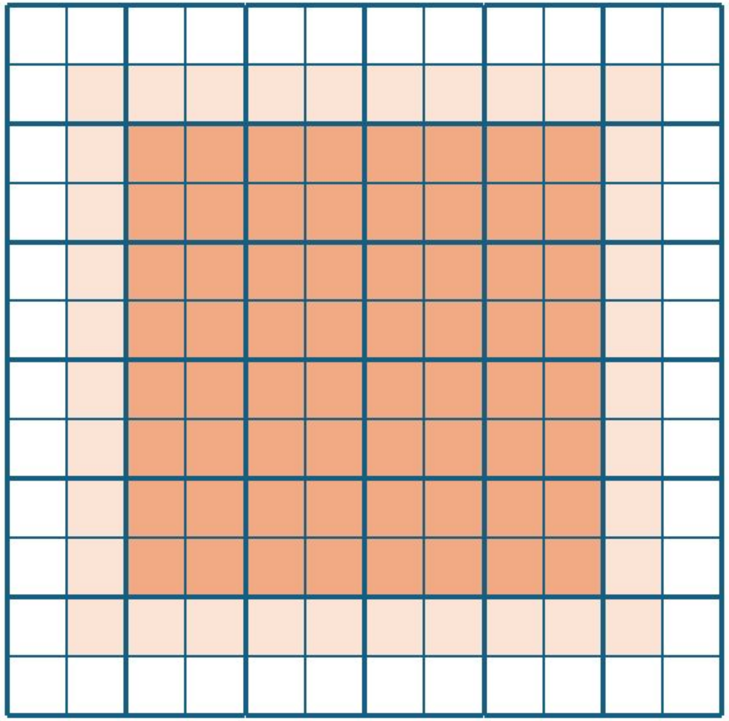
nsubgrid=1



nsubgrid=2

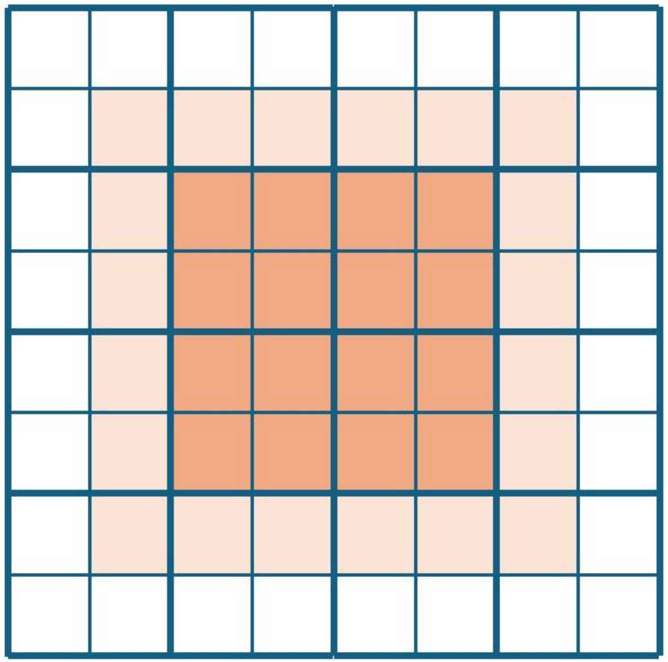


nsubgrid=4

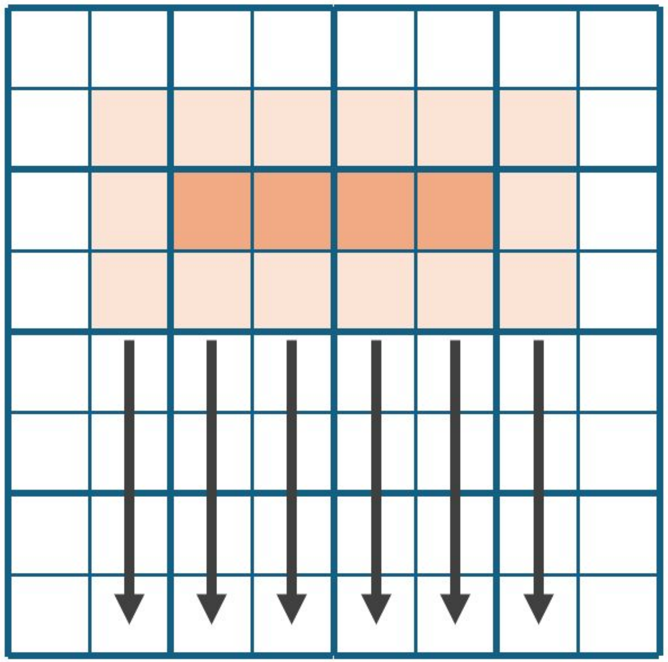


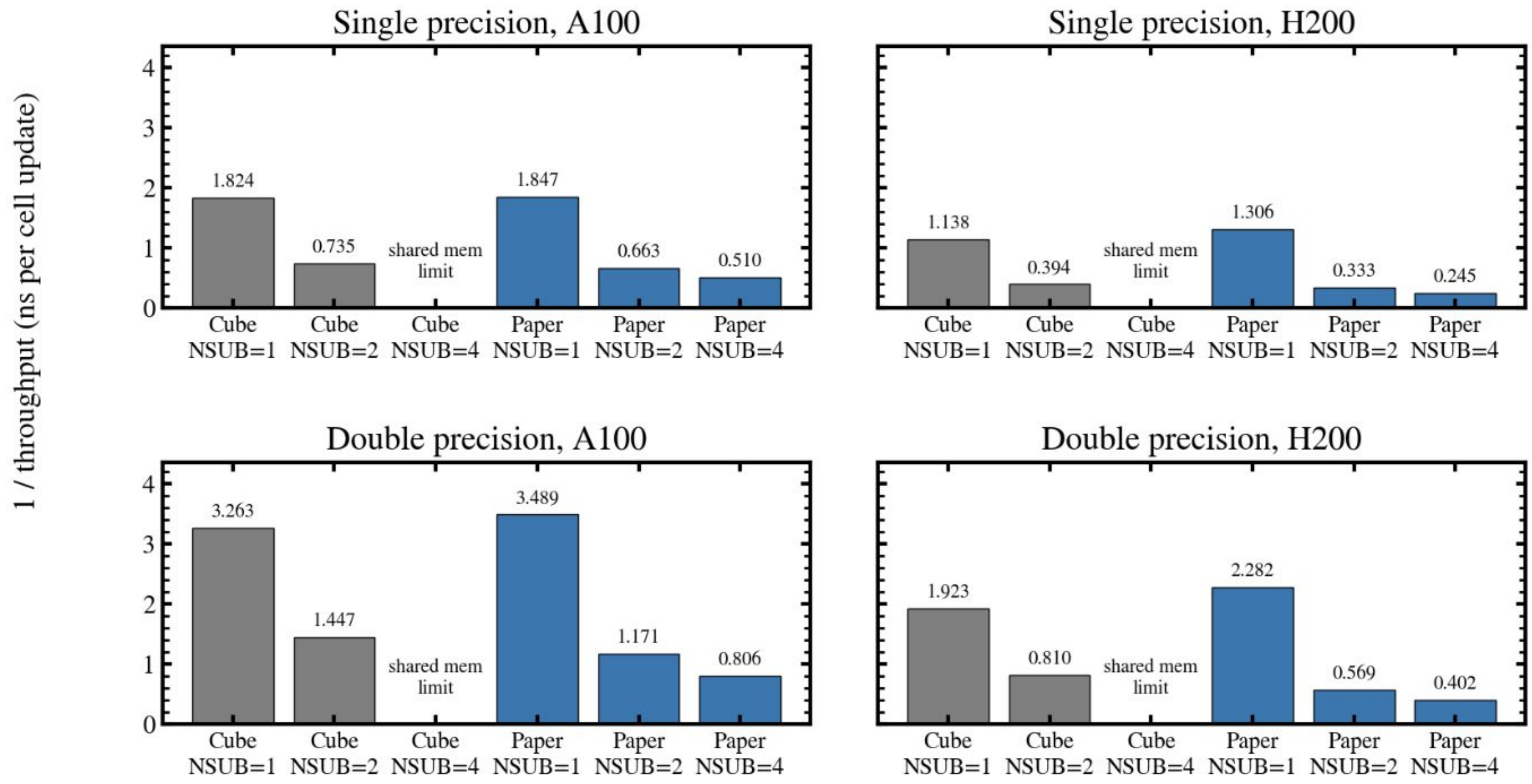
Optimizing shared memory with paper kernel

Cube kernel, nsubgrid=2, 216 cells

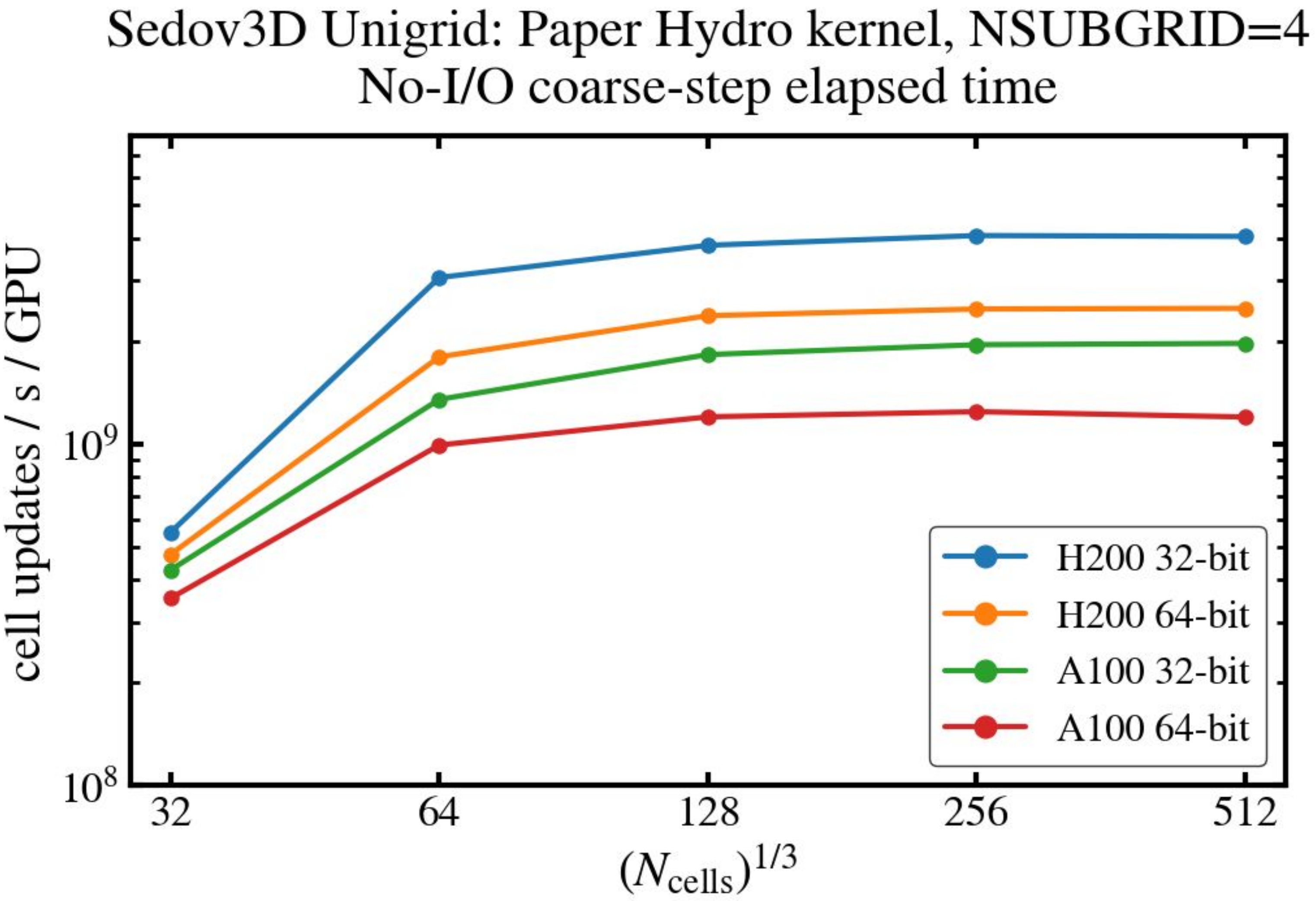


Paper kernel, nsubgrid=2, 108 cells



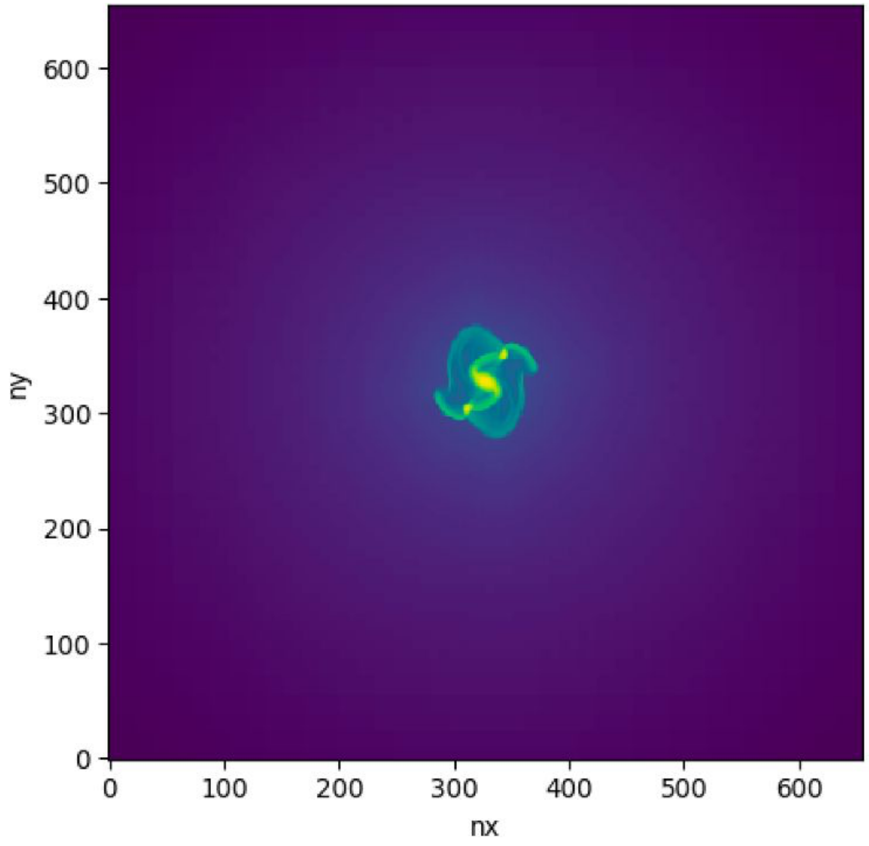


1 / throughput (ns per cell update)
Single precision, A100
1.824
0.735
shared mem limit
1.847
0.663
0.510
Single precision, H200
1.138
0.394
shared mem limit
1.306
0.333
0.245
Double precision, A100
3.263
1.447
shared mem limit
3.489
1.171
0.806
Double precision, H200
1.923
0.810
shared mem limit
2.282
0.569
0.402
Cube NSUB=1
Cube NSUB=2
Cube NSUB=4
Paper NSUB=1
Paper NSUB=2
Paper NSUB=4



OpenACC
More Science, Less Programming
OPEN HACKATHONS

GRAVITY SOLVER

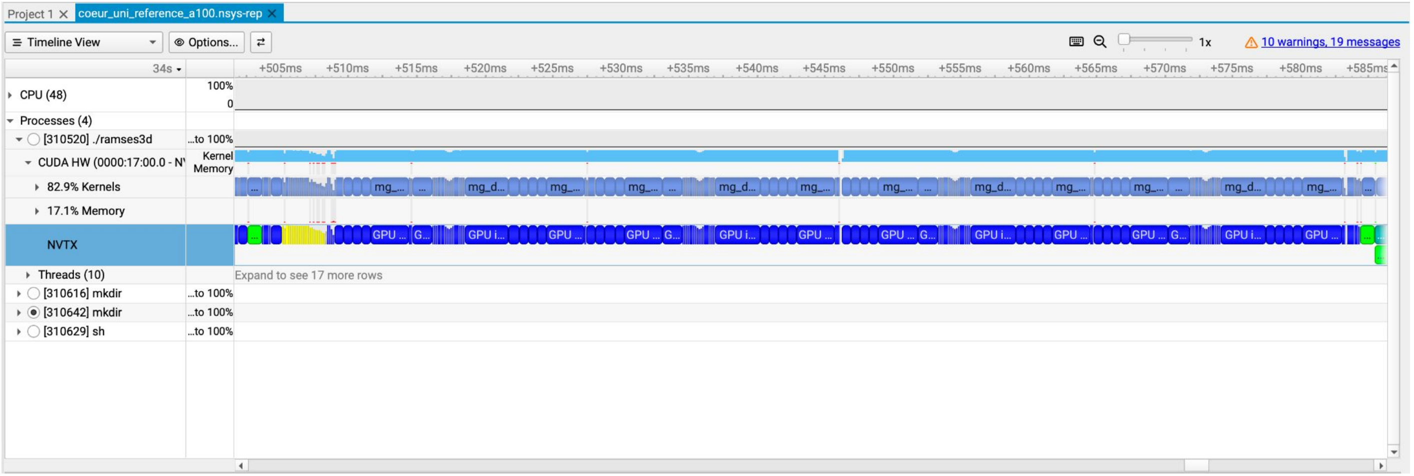


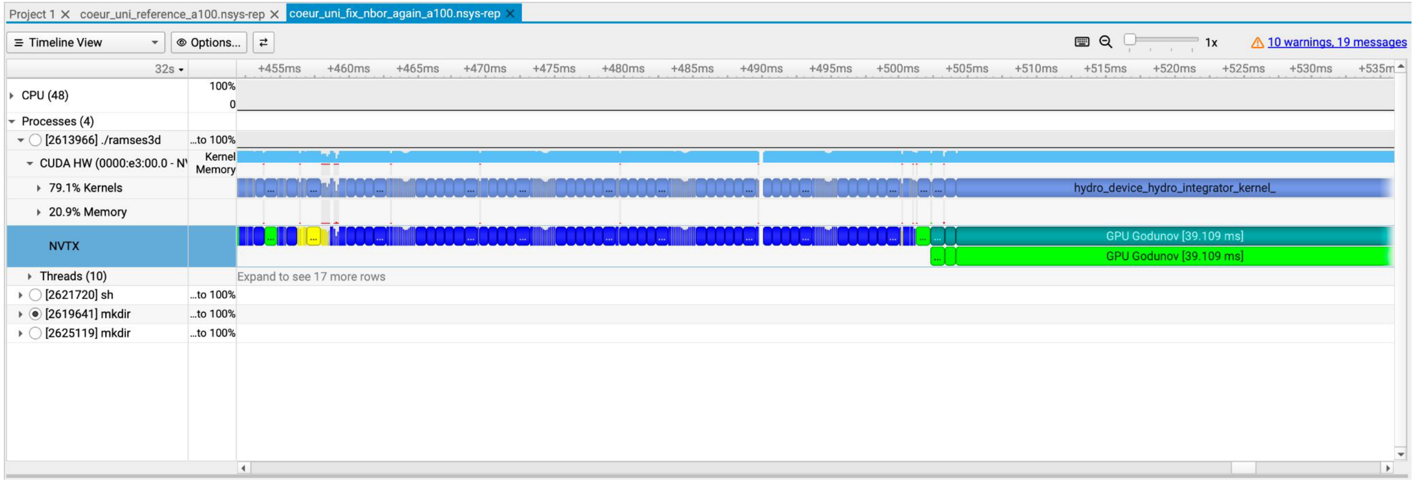
NSight Systems Output and Easy Optimizations

Minimized D2H and H2D memory copies

Fused kernels

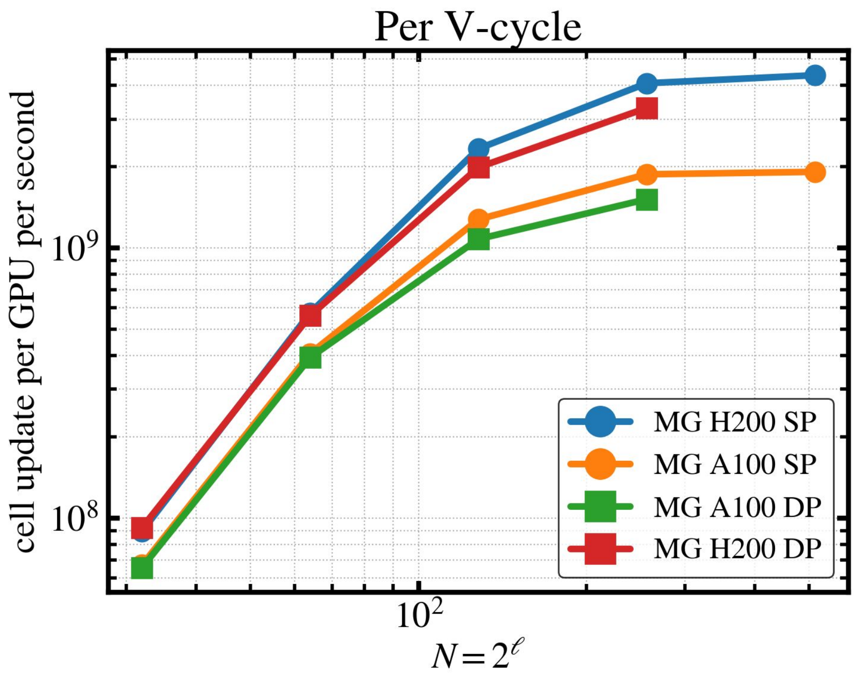
Minimized number of main GPU memory access



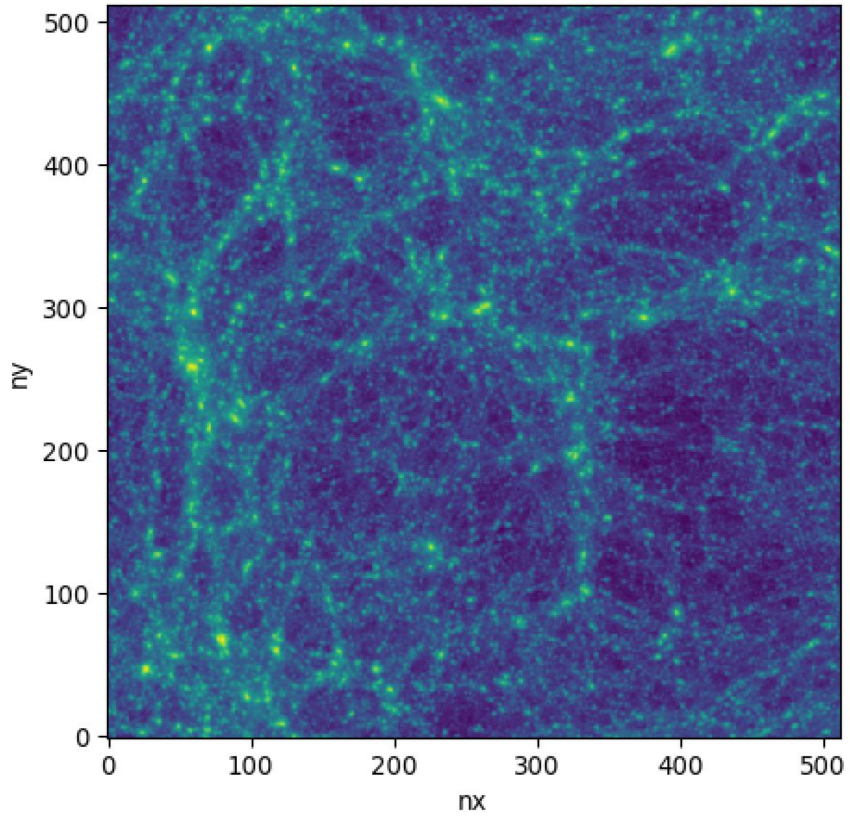


Progress and Goals

- We have profiled the code with NSight Systems
  - We have identified a few low hanging fruits
- We have benchmarked the gravity MG solver on A100 (H200) single precision
  - V-cycle: 2 (4) billion cell updates per GPU per second
  - MG hierarchy initialization: ½ V-cycle
- Squeeze more performance: fuse more kernels, avoid D2H copies



PARTICLE SOLVER

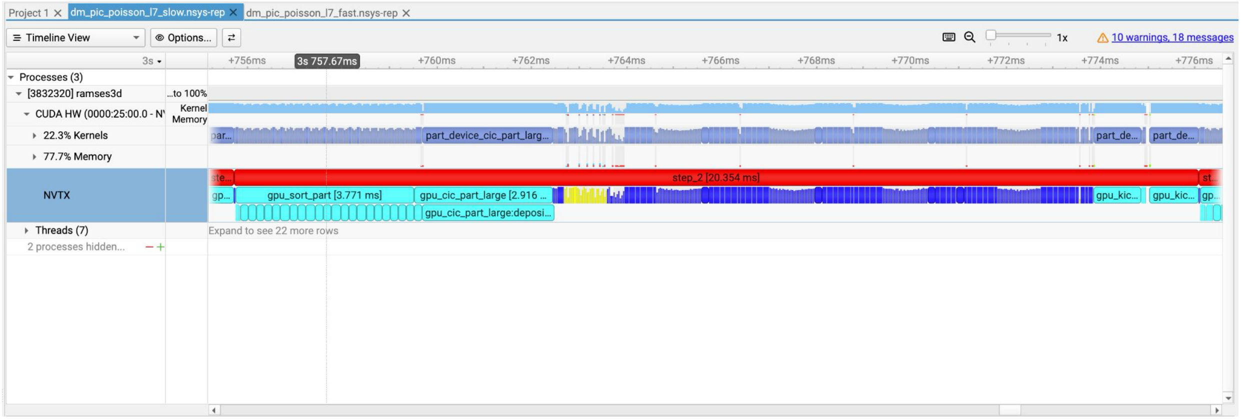


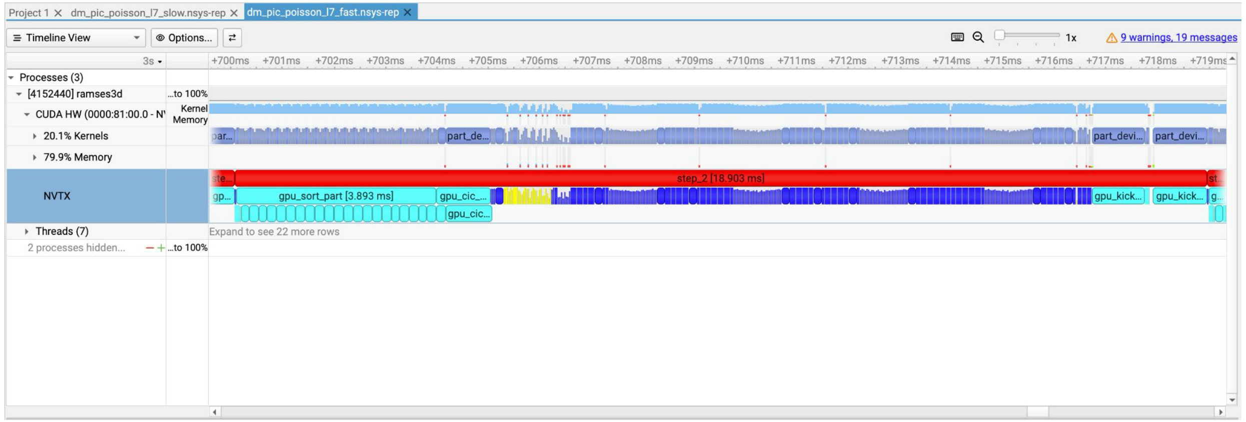
NSight Systems Reports

Particle Routines

Particle Routines

- Algorithm
  - Sort particles
  - Cloud-in-cell deposit particles onto grid
  - Multigrid to solve poisson's equation (gravity)
  - compute forces from gravitational potential and update positions & velocities
- Bottlenecks:
  - Our Radix Sort was slow…
  - CIC part was slow…





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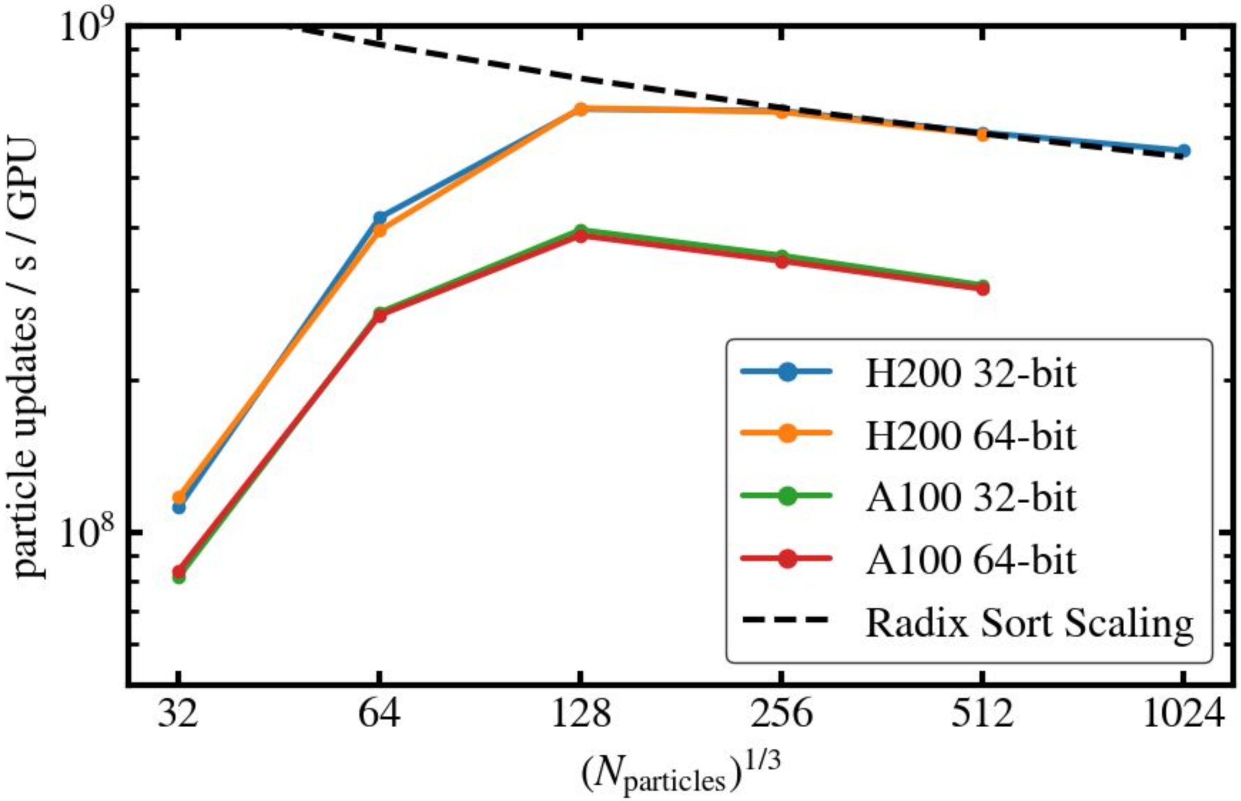
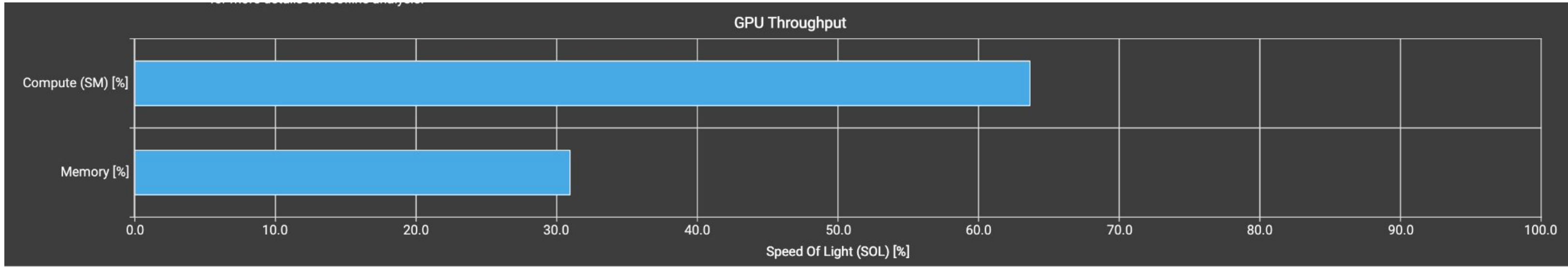
Performance Analysis

- Roofline Analysis
  - We see increase in performance as problem size increases to saturate GPU
  - once GPU saturates we see a decrease in performance (due to sort algorithm scaling)



Problems and Possible solutions



~~Atomic Add contentions~~

New CIC mass deposition scheme

Use CUB for sorting

Location	Value	Value (%)
gpu_hash.cuf:203 (0x152ef12b7180 in part_device_cic_part_warp_kernel_)	7,461,483	33
gpu_hash.cuf:202 (0x152ef12b6e60 in part_device_cic_part_warp_kernel_)	4,701,410	21
gpu_part.cuf:1173 (0x152ef12b7c70 in part_device_cic_part_warp_kernel_)	4,357,392	19
gpu_part.cuf:958 (0x152ef12b5b40 in part_device_cic_part_warp_kernel_)	970,611	4
gpu_part.cuf:958 (0x152ef12b5920 in part_device_cic_part_warp_kernel_)	970,611	4

All together: Isothermal hydro + DM + AMR